Randy Tang

rtang0203@gmail.com | 732-666-2746

SUMMARY

Software engineer with 3+ years building and optimizing high-throughput, production-critical data systems at a global bank. Strong CS/algorithms foundation (Cornell BS), deep Python proficiency, and hands-on fluency with agentic coding workflows and LLM APIs.

TECHNICAL SKILLS

Languages: Python, SQL, Java (fundamentals) | **Data/Systems:** Pandas, AMPS message queues, high-throughput pipelines, concurrency, Postgres, SQLite, Docker, Flask, FastAPI, BeautifulSoup, Google Places API
AI / Agentic: Agentic coding & orchestration, LLM APIs, tool-augmented workflows
Practices: Performance optimization, data pipeline design, web scraping & enrichment, code review, documentation & communication, Git

EXPERIENCE

Software Engineer — Bank of America Aug 2022 – Present

Cirrus OTC Trade Reporting | Python, Pandas, Quartz, AMPS message queues

- Cirrus trade reporting team — responsible for reporting all OTC trades through the bank.
- Designed, implemented, and improved high-throughput data pipelines processing tens of millions of transactions daily.
- Reference-data framework: pipeline of services via AMPS queues querying multiple external sources per trade event; handles retries, API calls, load balancing; extensible for new sources.
- T+1 UTI (Unique Trade Identifier) service: validates trade-ID uniqueness across counterparties, sends corrections; millions of transactions/day.
- High-volume backreporting service: parallelized, automated a formerly manual process, large speed/throughput gains.
- Transformer service (internal model -> outgoing XML) + DTCC response parser to update transaction status.
- Optimized services to improve speed/throughput/reliability by orders of magnitude.
- Python/Pandas scripts to scrape and analyze 50-100GB datasets.

Data Analyst (Contract) — Microsoft Mar 2022 – Aug 2022

Viva Topics | Python, Pandas, Jupyter

- Supported data science team on Viva Topics; processed/analyzed data used to train ML models.
- Wrote Python scripts to clean, annotate, and generate reports on user- and tester-collected data.

Engineering Intern — Northrop Grumman Fall 2019, Summer 2020

NX, ANSYS

- Released 50+ engineering drawings incl. new designs and large assemblies; fully designed/drafted/released multiple flight and test components.
- Structural analyses for large assemblies and detailed fastener-stack analyses.

PROJECTS

- **Med-Spa Lead-Gen Pipeline** (Python, SQLite, Google Places API, Anthropic API, BeautifulSoup) — Batch pipeline that discovers med-spas across metros via Google Places, enriches each lead's web presence (booking widgets, social, SSL, mobile), scores deficiency signals, generates personalized outreach hooks via Claude, and exports a ranked review queue. 8 idempotent stages, 20 tests, mock mode for offline dev.
- **Polymarket Scanner** (Flask, Discord API, SQLite) — Scans Polymarket for large trades and unusual/potential insider activity; alerts to Discord.
- **Market Data Dashboard** (Grafana, Twilio, Postgres, Docker) — Pulls market data (OI, funding, etc.) from Binance and Hyperliquid APIs; dashboard + SMS alerts on signal triggers.
- **Reading Recommendations App** (Flask, Gemini API, SQLite) — LLM-powered daily recommender over a Project Gutenberg corpus from user preferences.

EDUCATION

Cornell University, College of Engineering — Ithaca, NY Sep 2017 – Dec 2021

B.S. Computer Science & B.S. Mechanical Engineering | GPA 3.7

- CUAIR (Cornell Unmanned Aircraft) — autonomous plane with obstacle avoidance, object classification, airdrop; consistently top team at AUVSI SUAS competition.